

BDM PROJECT

VIVA REPORT

DATA-DRIVEN OPTIMIZATION IN BOTTLE MANUFACTURING

Name : Deepanshu Kamboj

Email : 23f2004097@ds.study.iitm.ac.in

Data Source : Primary

ORGANIZATION BACKGROUND

Business Name: SP Enterprises

Owner Name: Pankaj Kumar

Type : B2B

Established : 3 years ago

GST No. : 05AETFS4444M1ZC

Location : Bhagwanpur, Haridwar, -247661

Working Time: 09:00AM - 06:00PM



Problem Statements

1. Inefficient Raw Material Procurement Strategy

Sometimes materials are purchased in large quantity than needed

2. Demand-Supply Imbalance and Production Bottlenecks

Sometimes orders from clients are more than expected

3. Space Constraints

Above two problems are the reasons for space constraint in company

Data Collection & Metadata

Column Name	Description	Data Type
Date	Dates in which raw material was purchased	Date
Material	Name of Material purchased	Text
Supplier	Name of company who supplied raw material	Text
Quantity_kg	Quantity of raw material purchased	Number
Price_per_kg	Price of 1 kg of raw material	Number
Total Price	Total Price of Quantity purchased	Number

Raw material data

Source : Handwritten Bills

Duration : 5 months (Jan - May)

Data Size : 6 columns , 102 rows

Variables : Date, Material, Supplier

Quantity

Column Name	Description	Type
Date	Date on which order was placed	Date
Client	Name of company who gave order	Text
Bottle_Size_ml	The size of bottles which were ordered	Number
No_of_Bottles	Number of bottles were ordered	Number

Client Order Data

Source : Handwritten Bills

Duration : 5 months (Jan - May)

Data Size : 4 columns , 297 rows

Variables : Date, Client, Bottle_Size,

No. of bottles

Methodology

Trend Analysis

- Visualized raw material purchased over months.
- 2 per moving average was used for HDPE.

Correlation Analysis

- Relationship between raw material purchased and number of bottles ordered.
- Pearson Correlation was used.

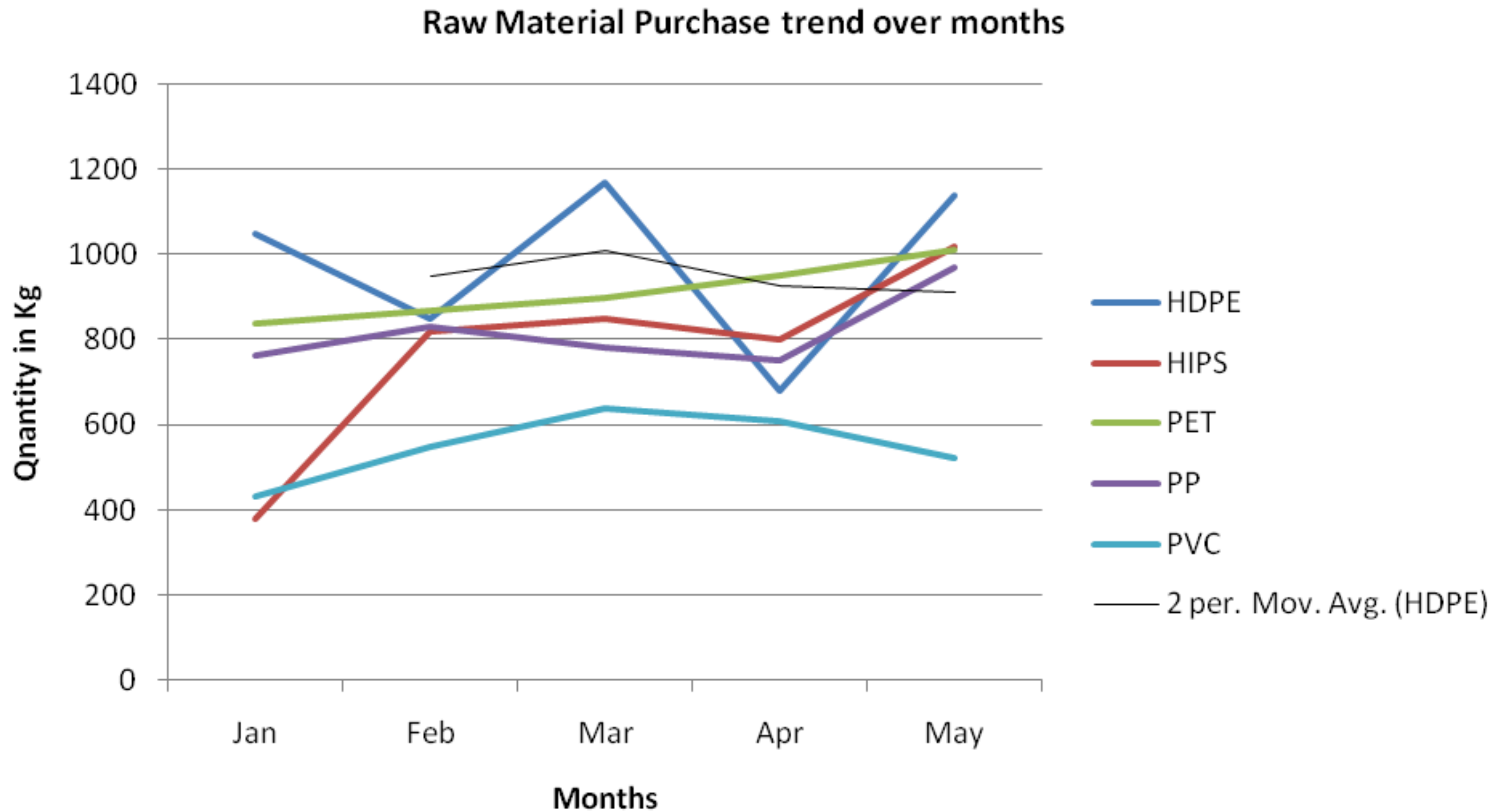
ABC Segmentation

- Ranked raw materials and clients by cumulative sales contribution.
- Categorized into A(top 70%), B(next 20%), C(bottom 10%).

Demand Forecasting

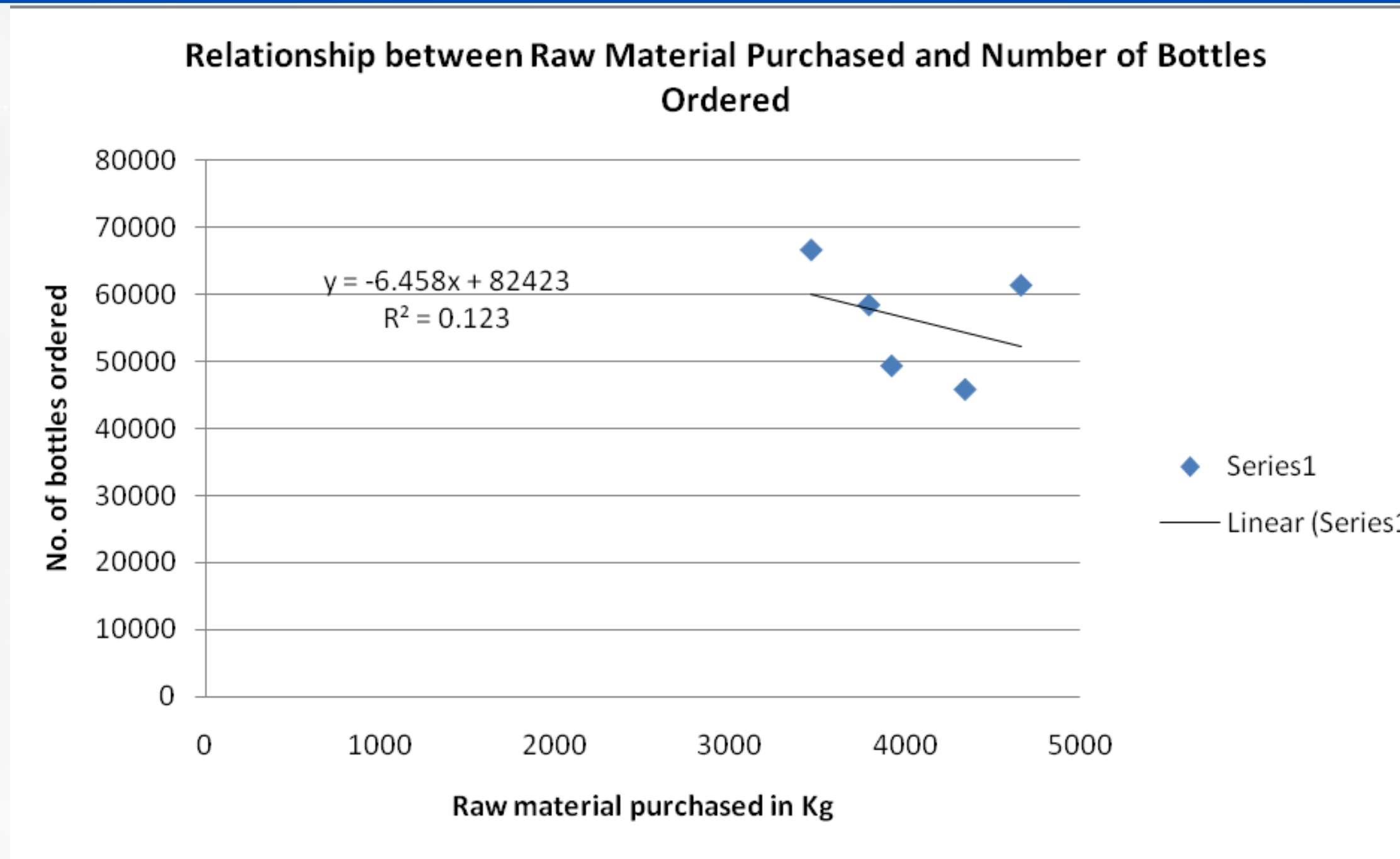
- Predict demand of bottles for next 3 months.
- Use 3-month moving average.

Trend Analysis



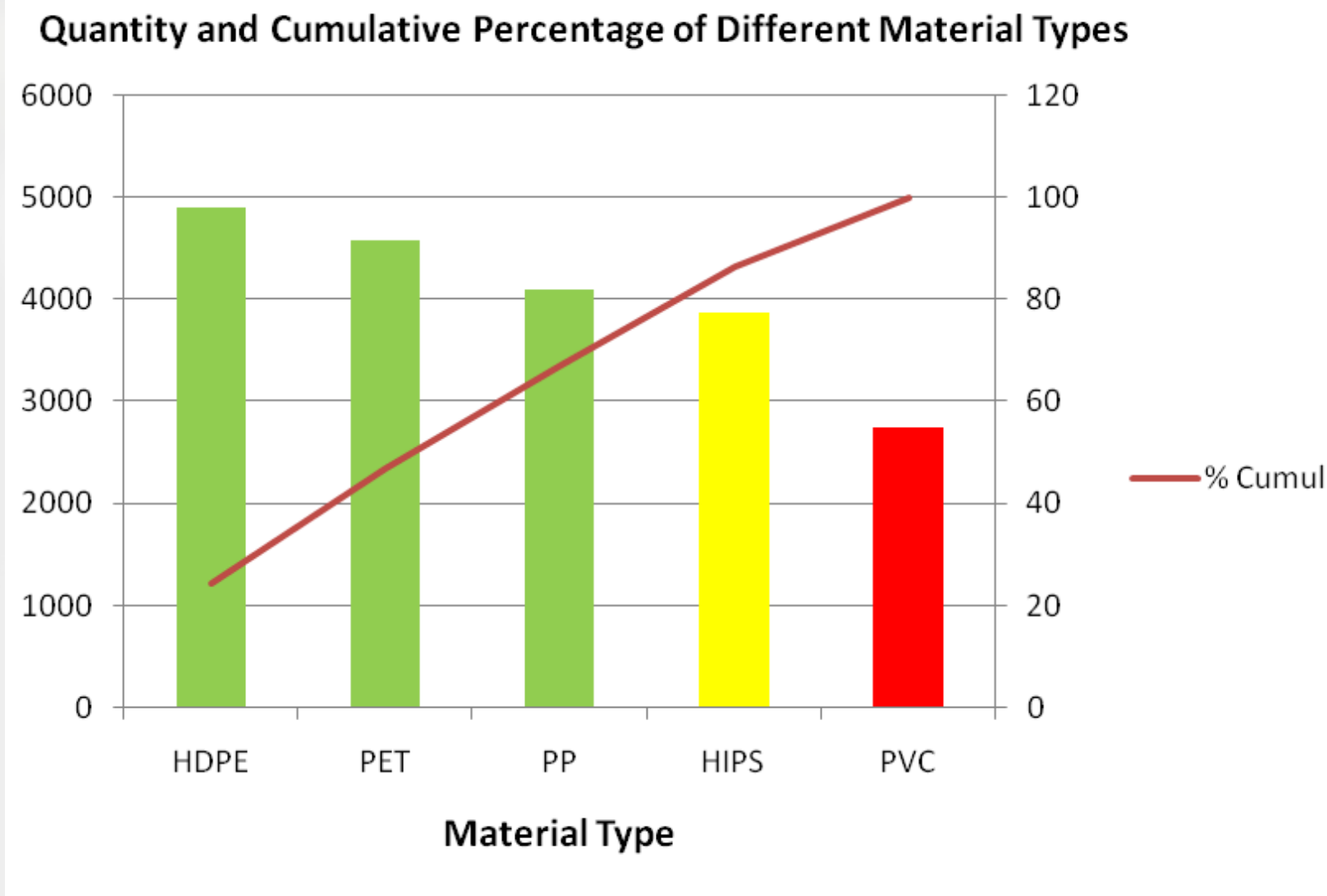
There is no clear plan or method being followed, which causes overstock or understock.

Correlation Analysis

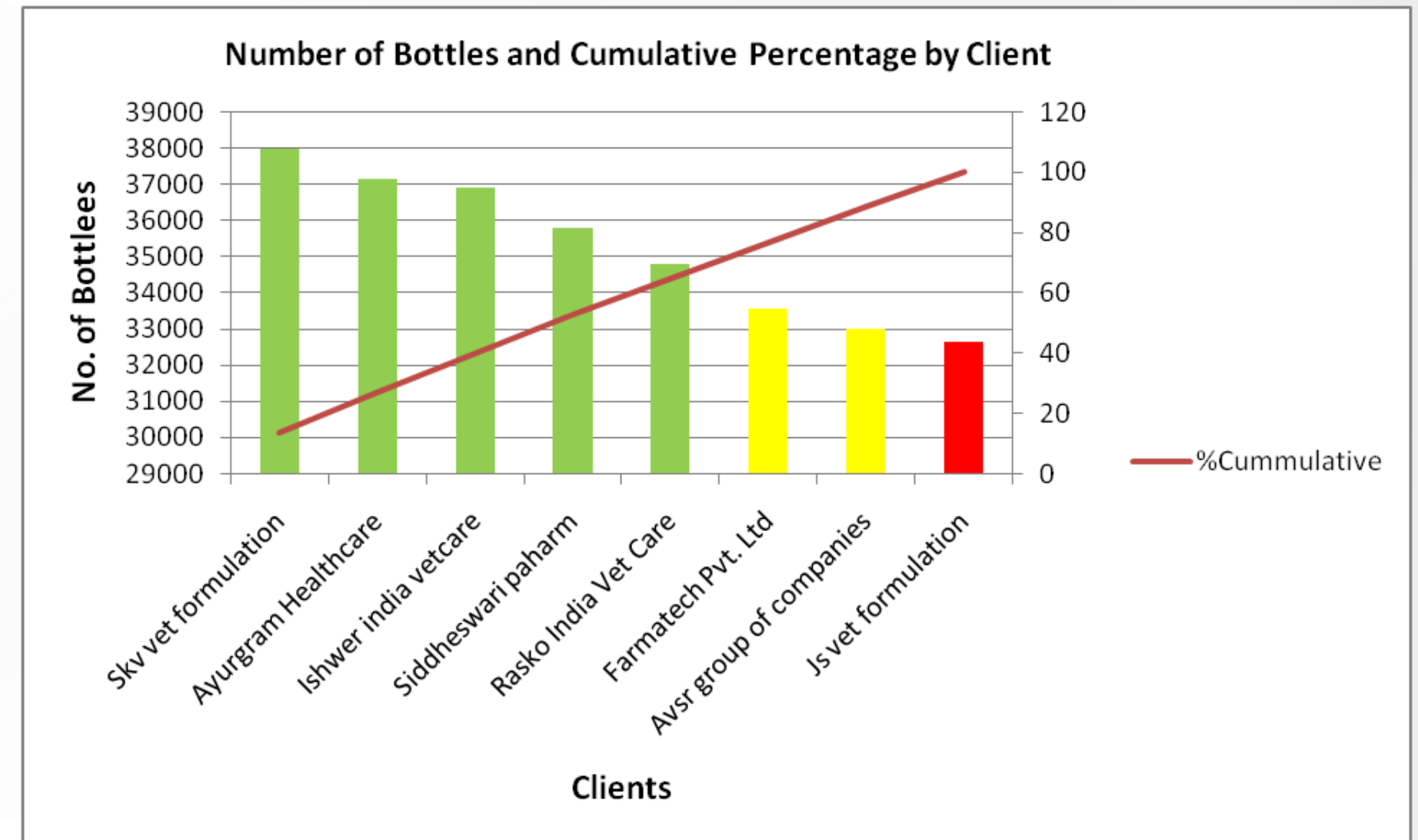


The correlation value is -0.35, which shows a moderate negative relationship

ABC Segmentation

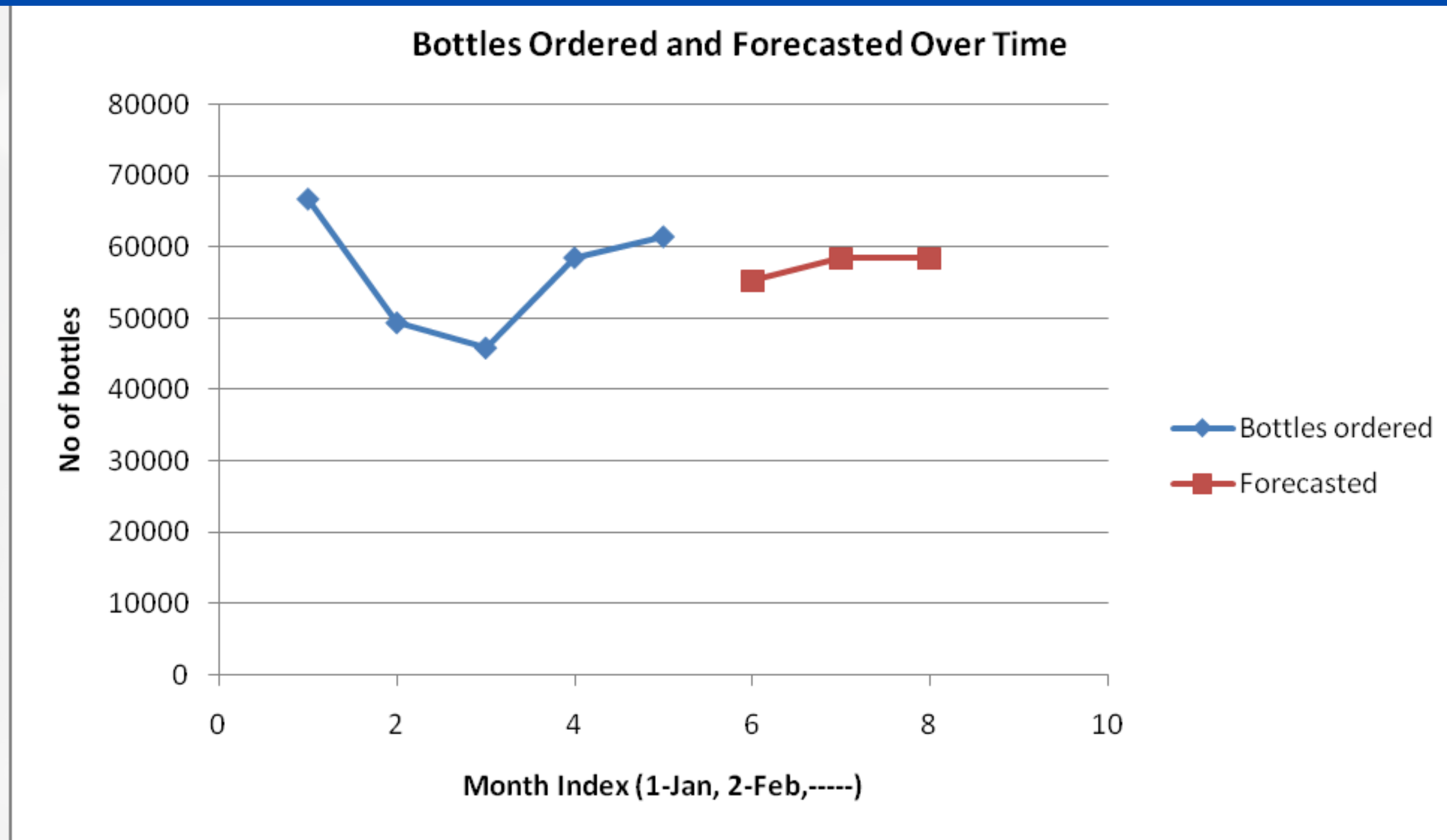


Category A - HDPE, PET, PP
Category B - HIPS
Category C - PVC



Category A - Skv, Ayurgram, Ishwer, Rasko, Siddeshwari
Category B - Farmatech, AVSR
Category C - Js Vet

Demand Forecast



The forecasted demand for bottles is expected to be steady around 55,000 to 59,000 bottles in the coming months

Recommendations

1. Focus more on A category raw materials and clients.
2. Don't overstock category C raw material.
3. Use forecast to prepare better in future.
4. Build strong relationship with top clients.
5. Explore why some materials or clients have low activity.

Thank You